1. Which specialized laboratory in an electrical engineering department is dedicated to testing motors, generators, and transformers?

(A) Electrical Machines Laboratory
(B) Digital Communications Lab
(C) Fluid Mechanics Lab
(D) Surveying Lab

2. What equipment is used in the Electrical Machines Lab to mechanically load an electric motor and measure its developed torque?

(A) Dynamometer (or Mechanical Brake Drum assembly)
(B) Manometer
(C) Thermocouple
(D) Barometer

3. What machine set is commonly used to demonstrate AC power generation and synchronization in the machines lab?

(A) DC Motor coupled to a 3-phase Synchronous Alternator (M-G Set)
(B) Diesel tractor engine
(C) Windmill model only
(D) Steam turbine boiler

4. What laboratory test is conducted to determine the dielectric breakdown voltage (BDV) of transformer mineral insulating oil?

(A) Oil Breakdown Voltage (BDV) Oil Test using standard cup and spherical/mushroom electrodes
(B) Litmus paper test
(C) Boiling point test
(D) Viscosity meter test

5. According to IS 335 / IEC 60156 standards, what is the minimum acceptable breakdown voltage for new filtered transformer oil?

(A) At least 30 kV (or 60 kV for EHV transformers across 2.5 mm gap)
(B) 12 V
(C) 230 V
(D) 100 V

6. Which specialized high-voltage apparatus is used to calibrate high peak voltages by sparkover distance in air?

(A) Sphere-Gap Voltmeter
(B) Digital Multimeter
(C) Wheatstone Bridge
(D) Potentiometer

7. What laboratory bench simulates long overhead AC transmission lines using lumped-pi and lumped-T ladder networks?

(A) Transmission Line Simulator / Artificial Line Bench
(B) Wind Tunnel
(C) Torsion Bench
(D) Tensile Testing Machine

8. What power electronic circuit in the drives lab converts constant 50 Hz AC utility voltage into variable DC voltage using Thyristors?

(A) Phase-Controlled Rectifier (SCR Converter)
(B) Autotransformer
(C) Rotary Spark Gap
(D) Linear Regulator

9. What device converts fixed DC voltage into variable DC voltage in power electronics laboratory experiments?

(A) DC Chopper (Buck / Boost Converter)
(B) Rotary Inverter
(C) Induction Coil
(D) Selenium Stack

10. What device converts DC power into variable-frequency AC power in the Power Electronics & Drives Lab?

(A) Inverter (e.g. 3-Phase Voltage Source Inverter)
(B) Rectifier
(C) Cycloconverter
(D) Step-up Transformer

11. Which transducer used in the instrumentation lab converts small linear mechanical displacement into an electrical AC voltage?

(A) LVDT (Linear Variable Differential Transformer)
(B) Strain Gauge
(C) Tachometer
(D) Bimetallic Strip

12. What resistance temperature detector (RTD) made of platinum with 100 ohms resistance at 0 deg C is standard in process control labs?

(A) Pt100 RTD
(B) Ni50
(C) Cu10
(D) Iron-Constantan

13. What temperature sensor operates on the Seebeck thermoelectric effect generated at the junction of two dissimilar metals?

(A) Thermocouple
(B) Thermistor
(C) RTD
(D) Pyrometer

14. Which laboratory bench trains students on industrial ladder logic programming, digital I/O cards, and relay interfaces?

(A) PLC & Industrial Automation Training Bench
(B) Soil Mechanics Bench
(C) Foundry Furnace
(D) Lathe Machine

15. What rotary electromagnetic transducer provides electrical signals proportional to shaft angular position for synchro-control transmitters?

(A) Synchro / Resolver
(B) Stepper Motor
(C) Permanent Magnet
(D) Solenoid

16. What is the premier global professional technical society for electrical and electronics engineers?

(A) IEEE (Institute of Electrical and Electronics Engineers)
(B) ASME
(C) ASCE
(D) AIChE

17. What is the dedicated IEEE society focused on electric power generation, transmission, distribution, and smart grids?

(A) IEEE Power & Energy Society (IEEE PES)
(B) IEEE Computer Society
(C) IEEE Photonics Society
(D) IEEE Robotics Society

18. Which IEEE society focuses on industrial electrical applications, motor drives, factory systems, and mining electrification?

(A) IEEE Industry Applications Society (IEEE IAS)
(B) IEEE Solid-State Circuits Society
(C) IEEE Antennas Society
(D) IEEE Oceanic Society

19. Which IEEE society specializes in power semiconductor devices, converters, inverters, and switch-mode supplies?

(A) IEEE Power Electronics Society (IEEE PELS)
(B) IEEE Magnetics Society
(C) IEEE Geoscience Society
(D) IEEE Aerospace Society

20. What is the global affinity group within IEEE dedicated to promoting women engineers and scientists?

(A) IEEE Women in Engineering (IEEE WIE)
(B) IEEE Young Professionals
(C) IEEE Life Members
(D) IEEE Consultants

21. What apex statutory chartered engineering organization in India was established in 1920 to advance engineering professions?

(A) The Institution of Engineers (India) - IEI
(B) Bar Council of India
(C) Medical Council
(D) Press Trust

22. What UK-based global professional engineering institution awards Chartered Engineer (CEng) credentials?

(A) The Institution of Engineering and Technology (IET)
(B) IEEE
(C) SAE
(D) CIE

23. What is the primary role of an active IEEE Student Branch at a university campus?

(A) Organizing technical workshops, expert guest lectures, industrial visits, hackathons, and networking with industry mentors
(B) Selling textbooks
(C) Managing campus cafeteria
(D) Printing exam admit cards

24. What safety acronym represents the procedure of isolating hazardous energy sources and locking breakers before maintenance?

(A) LOTO (Lockout / Tagout)
(B) FIFO
(C) SCADA
(D) HVDC

25. What standard color is universally associated with Danger and High Voltage Warning signage?

(A) Red and Yellow with Warning Triangle
(B) Light Blue
(C) Pastel Green
(D) Light Pink

26. What safety floor matting must be placed in front of high-voltage electrical panels according to IS 15652 standards?

(A) Dielectric Insulating Rubber Mats
(B) Bare Steel Plate
(C) Wet Cotton Rug
(D) Carpet Tile

27. What fire extinguisher class is strictly mandatory for electrical fires involving energized equipment?

(A) Class C / Carbon Dioxide (CO_2) or Clean Agent Halotron Extinguisher
(B) Plain Pressurized Water Canister
(C) Aqueous Film Foam
(D) Wet Chemical Kitchen

28. Why must WATER NEVER be sprayed onto energized electrical equipment fires?

(A) Water conducts electricity, creating a lethal electrocution path back to the firefighter and triggering violent short-circuit explosions
(B) Water increases room temperature
(C) Water damages plastic labels
(D) Water makes noise

29. What primary first-aid procedure should be immediately administered if an electrocution victim is unconscious and not breathing?

- **(A)** Cardiopulmonary Resuscitation (CPR) and deploy Automated External Defibrillator (AED)
- **(B)** Give cold water to drink
- **(C)** Cover with wet blankets
- **(D)** Leave in place for an hour

30. Before touching any exposed electrical circuit in a laboratory, what test procedure must always be executed?

- **(A)** Verify de-energized state using a certified voltage tester (Live-Dead-Live test rule)
- **(B)** Touch with finger quickly
- **(C)** Assume it is off if silent
- **(D)** Check main clock

31. What emergency mechanical button is installed prominently in every electrical laboratory to immediately cut all power?

- **(A)** Emergency Stop (E-Stop) Mushroom Pushbutton
- **(B)** Reset Switch
- **(C)** Buzzer Switch
- **(D)** Light Dimmer

32. What personal protective item must be worn by all students and technicians in the high-voltage laboratory at all times?

- **(A)** Safety Glasses / Face Shield and Insulating Safety Shoes
- **(B)** Loose ties and scarves
- **(C)** Gold bracelets
- **(D)** Open-toe sandals

33. What grounding tool is used to discharge residual capacitive charges from de-energized high-voltage test cables?

- **(A)** Grounding Discharge Stick (Earthing Rod with resistor)
- **(B)** Wooden Broom
- **(C)** Plastic Ruler
- **(D)** Steel Crowbar

34. What is the statutory electrical safety regulation body in India operating under the Ministry of Power?

- **(A)** Central Electricity Authority (CEA) Regulations
- **(B)** SEBI
- **(C)** IRDAI
- **(D)** TRAI

35. What is the minimum attendance requirement mandated by university academic policy at LPU to be eligible for end-term examinations?

- **(A)** 75% minimum attendance
- **(B)** 25%
- **(C)** 50%
- **(D)** 10%

36. In university continuous assessment (CA), what does 'Formative Assessment' aim to achieve?

- **(A)** Ongoing feedback during the learning process through quizzes, lab assignments, and mini-projects to improve student mastery
- **(B)** Final terminal grading only
- **(C)** Punishing absent students
- **(D)** Ranking colleges

37. What academic policy prohibits unauthorized copying, fabrication of experimental lab data, or using uncredited sources?

- **(A)** Academic Integrity and Anti-Plagiarism Policy
- **(B)** Library Loan Policy
- **(C)** Hostel Curfew Policy
- **(D)** Uniform Dress Code

38. In engineering laboratory evaluation, what is a 'Continuous Evaluation Rubric'?

- **(A)** A standardized scoring criteria matrix evaluating pre-lab preparation, circuit wiring, data observation, analysis, and viva-voce
- **(B)** A random number generator
- **(C)** A written penalty notice
- **(D)** A grade curve

39. What is the culminating two-semester team project undertaken by senior undergraduate engineering students called?

- **(A)** Capstone Design Project
- **(B)** Weekly Homework
- **(C)** Freshman Quiz
- **(D)** Mid-term Essay

40. What is the defining purpose of integrating industrial safety standards into university electrical engineering curricula?

- **(A)** Cultivating safety-first professional engineering culture that prevents industrial fatalities and protects public infrastructure
- **(B)** Extending lecture durations
- **(C)** Fulfilling paperwork bureaucracy only
- **(D)** Increasing textbook sales

41. In the Electrical Machines Lab, why is a DC Shunt Motor NEVER started directly across full line voltage without a 3-point or 4-point starter?

- **(A)** At standstill ($N = 0$), back-EMF is zero ($E_b = 0$), causing enormous starting current ($I_a = V / R_a$) that can burn windings and trip breakers
- **(B)** Motor will run in reverse
- **(C)** Field flux becomes zero
- **(D)** Motor cannot stop

42. What dangerous condition occurs if the field winding of an unloaded DC Shunt Motor accidentally opens during operation?

- **(A)** Field flux collapses to residual values ($\Phi \to \Phi_{res}$), causing armature speed to accelerate uncontrollably to destructive runaway speeds ($N \propto 1/\Phi$)
- **(B)** Motor immediately stops
- **(C)** Voltage becomes AC
- **(D)** Armature locks

43. In synchronous alternator testing, what does the Open-Circuit Characteristic (OCC) plot?

- **(A)** Generated open-circuit terminal voltage V_{oc} versus DC field excitation current I_f at rated synchronous speed
- **(B)** Armature current vs speed
- **(C)** Power factor vs torque
- **(D)** Core loss vs frequency

44. What does the Short-Circuit Characteristic (SCC) of a synchronous generator plot?

- **(A)** Sustained short-circuit armature current I_{sc} versus DC field excitation current I_f with terminals shorted
- **(B)** Terminal voltage vs load
- **(C)** Rotor temperature vs slip
- **(D)** Power factor vs angle

45. How is the unsaturated Synchronous Impedance (Z_s) of an alternator calculated from OCC and SCC data?

- **(A)** $Z_s = \frac{V_{oc} \text{ (from air-gap line)}}{I_{sc}} \text{ at the same field current } I_f$
- **(B)** $Z_s = V_{oc} \cdot I_{sc}$
- **(C)** $Z_s = V_{oc} + I_{sc}$
- **(D)** $Z_s = I_{sc} / V_{oc}$

46. What is the Short-Circuit Ratio (SCR) of a synchronous generator, and what does a high SCR indicate?

- **(A)** Ratio of field current for rated open-circuit voltage to field current for rated short-circuit current; high SCR indicates large air gap, superior voltage regulation, and higher stability margins
- **(B)** Ratio of resistance to reactance
- **(C)** Transformer turn ratio
- **(D)** Acoustic noise ratio

47. In transformer testing, what is the 'Sumpner's Test' (Back-to-Back Test) uniquely used for?

- **(A)** Tests full-load temperature rise and efficiency of two identical transformers under full load conditions without drawing full load power from the mains
- **(B)** Measures winding turns
- **(C)** Tests bushing oil breakdown
- **(D)** Measures core weight

48. In the Power Systems Lab, what does the 'Three-Lamp Method' or 'Synchroscope' verify before synchronizing an alternator to the busbar?

- **(A)** Verifies that voltage magnitude, frequency, phase sequence, and phase angle of incoming alternator match the grid busbar exactly
- **(B)** Tests light bulb durability
- **(C)** Measures room illumination
- **(D)** Tests neutral earthing

49. What happens if a circuit breaker is closed to synchronize an alternator when its phase angle is 180° out of phase with the grid?

- **(A)** A violent short circuit of twice the system voltage develops, generating catastrophic electromagnetic forces that can twist and shear generator shafts
- **(B)** The generator runs smoothly
- **(C)** Power factor becomes unity
- **(D)** Frequency doubles peacefully

50. What is the function of a Buchholz Relay installed on large oil-immersed power transformers?

- **(A)** Gas-actuated protective relay installed in the pipe between transformer tank and conservator to detect internal slow gas generation (fault warning) and oil surges (instant trip)
- **(B)** Pumps cooling oil
- **(C)** Measures ambient temperature
- **(D)** Filters sludge

51. What does the Silica Gel Breather on a transformer conservator do, and what does a color change from blue to pink signify?

- **(A)** Dehydrates incoming air; color change from cobalt blue to pink indicates that the silica gel is saturated with moisture and requires replacement/re-baking
- **(B)** Indicates high voltage
- **(C)** Indicates transformer overload
- **(D)** Measures core temperature

52. In high-voltage laboratories, what is a Marx Generator circuit used to synthesize?

- **(A)** High-voltage lightning impulse surges by charging capacitors in parallel via resistors and discharging them in series through spark gaps
- **(B)** Continuous DC power
- **(C)** Sinusoidal AC power
- **(D)** Radio frequency signals

53. What is the standard lightning impulse waveform specification standardized under IEC 60060?

- **(A)** $1.2 / 50\,\mu\text{s}$ wave (Front time $T_1 = 1.2\,\mu\text{s} \pm 30\%$, Tail time $T_2 = 50\,\mu\text{s} \pm 20\%$)
- **(B)** $10 / 1000\,\mu\text{s}$
- **(C)** $50 / 60\text{ Hz}$
- **(D)** $1\text{ ms} / 10\text{ ms}$

54. In the Power Electronics Lab, why is an optocoupler driver IC (e.g. TLP250 or HCPL-3120) inserted between a microcontroller PWM pin and an IGBT gate?

- **(A)** Provides high-voltage galvanic isolation ($> 2500\text{ V}$) and delivers high peak gate current to charge/discharge IGBT gate capacitance rapidly
- **(B)** Inverts logic polarity only
- **(C)** Generates PWM signals
- **(D)** Regulates DC bus voltage

55. What occurs during 'Commutation Failure' in a line-commutated thyristor inverter?

- **(A)** Incoming thyristor fails to take over current or outgoing thyristor fails to recover forward blocking capability before its anode voltage becomes positive, causing a short circuit across the DC link
- **(B)** Motor runs faster
- **(C)** Thyristor turns off forever
- **(D)** Frequency drops

56. What is the primary difference between an RTD and a Thermistor in temperature measurement labs?

- **(A)** RTDs use pure metals (platinum) with positive temperature coefficient (PTC) and high linearity; Thermistors use ceramic semiconductors with high sensitivity and non-linear negative temperature coefficient (NTC)
- **(B)** RTDs are non-linear
- **(C)** Thermistors use platinum
- **(D)** Thermistors cannot measure heat

57. In control engineering laboratories, what does an optical rotary encoder's 'Index Channel' (Z-pulse) provide?

(A) Exactly one reference pulse per complete 360° mechanical revolution to establish an absolute zero reference position
(B) Direction of rotation only
(C) Speed pulses only
(D) Clock signal

58. What does the IEEE PES Student Congress provide for undergraduate electrical engineering members?

(A) Global networking with power utility leaders, technical paper presentation tracks, leadership training, and career mentorship
(B) Subsidized laptop sales
(C) Free textbooks
(D) Housing scholarships

59. What is the role of an IEEE Student Branch Chapter Chair?

(A) Leading the student executive committee, organizing technical events, managing chapter finances, and reporting branch activities to IEEE headquarters
(B) Grading university exams
(C) Teaching lectures
(D) Enforcing hostel rules

60. What prestigious annual award is presented by the IEEE Power & Energy Society to outstanding undergraduate engineering students?

(A) IEEE PES Scholarship Plus Initiative
(B) Turing Award
(C) Nobel Prize
(D) Fields Medal

61. What does the Institution of Engineers (India) require for granting the professional title of 'Chartered Engineer (India)'?

(A) Accredited engineering degree, minimum years of verified professional engineering experience, and adherence to professional code of ethics
(B) Passing a driving test
(C) Paying a one-time fee only
(D) Publishing a novel

62. What is 'Lockout/Tagout' (LOTO) standardized under OSHA 29 CFR 1910.147?

(A) Placing physical padlocks on electrical circuit breakers and warning tags indicating that equipment is under maintenance and must not be energized
(B) Computer password protection
(C) Locking laboratory doors at night
(D) Tagging laboratory inventory

63. What is an 'Arc Flash Blast' and what creates its explosive mechanical pressure wave?

(A) Rapid superheating of air and vaporization of solid copper conductors (expanding 67,000 times in volume), creating explosive blast pressures exceeding 2000 lbs/sq ft and molten shrapnel
(B) Compressed air tank rupture
(C) Hydrogen combustion
(D) Chemical acid spill

64. According to NFPA 70E, what is the 'Limited Approach Boundary'?

(A) An approach boundary to exposed energized conductors within which a shock hazard exists, requiring visual warning and unqualified persons to be escorted by qualified personnel
(B) Building property line
(C) Distance from transformer fence
(D) Substation gate

65. What is the 'Restricted Approach Boundary' for shock protection?

(A) A distance from exposed energized conductors where only qualified electrical personnel with approved insulating PPE and insulated tools are permitted to enter
(B) Substation control room
(C) Parking lot boundary
(D) Public road edge

66. What safety rule is known as the 'One-Hand Rule' when working around energized high-voltage circuits?

(A) Keep one hand in your pocket or behind your back to prevent creating a current path across the chest and heart in case of accidental contact
(B) Work with one eye closed
(C) Use only one finger
(D) Hold tools with one hand only

67. In Indian Electricity Rules 1956, what is the mandatory earth loop resistance limit for a domestic installation?

(A) Not exceeding 1 to 2 Ohms (and not exceeding 0.5 to 1 Ohm for large generating substations)
(B) 1000 Ohms
(C) 100 Ohms
(D) 500 Ohms

68. What is a Residual Current Circuit Breaker (RCCB) / Earth Leakage Circuit Breaker (ELCB) designed to do?

(A) Monitors differential current between phase and neutral; if imbalance exceeding 30 mA occurs (indicating leakage to earth through human touch), it trips within 30 ms to prevent fatal ventricular fibrillation
(B) Protects against overloads
(C) Maintains constant voltage
(D) Filters power surges

69. In laboratory capstone projects, what is 'Peer Assessment' in team dynamics?

(A) Constructive evaluation of team members' technical contributions, punctuality, collaboration, and accountability according to clear rubric criteria
(B) Grading student quizzes
(C) Electing class representatives
(D) Ranking students by height

70. What is a 'Non-Disclosure Agreement' (NDA) in sponsored university-industry research projects?

(A) A legal contract committing researchers and students to keep confidential proprietary industrial trade secrets, designs, and patentable data
(B) A software license
(C) A student admission form
(D) An exam registration slip

71. What is 'Intellectual Property' (IP) generated during student capstone engineering research?

(A) Creations of the mind, including patents for novel circuits, copyrights for software code, and industrial design registrations
(B) Laboratory furniture
(C) Oscilloscopes and meters
(D) Textbook library books

72. What is a 'Patent' in electrical engineering innovation?

(A) An exclusive legal right granted by a government for a novel, non-obvious, and industrially applicable technical invention for a limited period (typically 20 years)
(B) A copyright on text
(C) A university diploma
(D) A publication certificate

73. In scholarly publishing, what constitutes 'Self-Plagiarism' (Text Recycling)?

(A) Reusing significant portions of one's own previously published papers without citation, misleading readers about the novelty of the work
(B) Quoting another author with citations
(C) Translating a textbook
(D) Reviewing a paper

74. What ethical mandate requires engineers to report unsafe conditions or defective designs to authorities if management fails to act?

(A) Whistleblowing obligation under the Professional Engineering Code of Ethics
(B) Insider trading
(C) Corporate espionage
(D) Public relations marketing

75. What is the primary benefit of achieving 'Best Paper Award' or presenting at prestigious conferences like IEEE PES General Meeting?

(A) Validation of technical research quality by international peer experts, career networking with premier research groups, and enhanced postgraduate scholarship prospects
(B) Monetary cash prizes only
(C) Exemption from all university exams
(D) Free flight tickets for life

76. In IEEE Std C57.12.00, what is the significance of the transformer impedance voltage (%Z) measured during a short-circuit test?

(A) It determines the internal voltage regulation under load, the prospective symmetrical short-circuit current during terminal faults ($I_{sc} = I_{rated} / \%Z$), and parallel load sharing between transformers
(B) It determines oil dielectric strength
(C) It measures core weight
(D) It dictates tank paint color

77. In high-voltage testing, how does atmospheric air density correction factor (δ) affect the sparkover voltage of sphere gaps?

(A) $\delta = \frac{0.386 P}{273 + T}$; sparkover voltage is directly proportional to δ, requiring empirical correction for barometric pressure P (mmHg) and temperature T ($^\circ\text{C}$)
(B) Sparkover is independent of air density
(C) Higher temperature increases sparkover
(D) Humidity has zero influence

78. What does the Polarization Index (PI) test evaluate in high-voltage machine stator insulation?

(A) Ratio of insulation resistance measured at 10 minutes to that at 1 minute (R_{10} / R_1); a PI > 2.0 indicates clean, dry, healthy insulation, while PI < 1.5 indicates moisture contamination and degradation
(B) Ratio of voltage to current
(C) Motor efficiency
(D) Winding resistance ratio

79. In partial discharge measurement (IEC 60270), what is 'Apparent Charge' measured in picocoulombs (pC)?

(A) The charge pulse that, if injected instantaneously across the terminals of the test object, would produce the same voltage transient as the internal partial discharge spark itself
(B) Total battery charge
(C) Stator current integral
(D) Capacitor static charge

80. What is the difference between Type Tests, Routine Tests, and Special Tests on electrical switchgear under IEC 62271?

(A) Type tests prove design compliance on prototype units (short-time withstand, temperature rise); Routine tests verify manufacturing quality on every single production unit; Special tests investigate specific client requirements
(B) Type tests are done on site; Routine tests are done in labs
(C) There is no difference
(D) Routine tests are done once a decade

81. What did the infamous 1986 Chernobyl disaster and 2003 North American Blackout highlight regarding engineering human factors and safety culture?

(A) Disabling safety interlocks for unauthorized experiments, ignoring alarms, and inadequate operator situational awareness can trigger catastrophic grid collapse and loss of life
(B) Computers should be banned
(C) AC power is dangerous
(D) Relays are unnecessary

82. In the Challenger Space Shuttle disaster (1986), what ethical breakdown occurred between engineering recommendations and management pressure?

(A) Engineers warned that cold temperatures would cause O-ring seal failure, but management pressured an override to meet launch schedules, demonstrating the peril of compromising safety for operational convenience
(B) A software bug occurred
(C) An electrical wire shorted
(D) A fuel tank was empty

83. What does the 'Precautionary Principle' dictate in environmental and public health engineering?

(A) Where there are threats of serious or irreversible damage, lack of full scientific certainty shall not be used as a reason for postponing cost-effective measures to prevent environmental degradation
(B) Ignore risks until accidents occur
(C) Never construct power plants
(D) Prioritize immediate profits

84. Under the IEEE Code of Ethics, how must an engineer resolve a conflict of interest between their employer and an outside consulting client?

(A) Promptly and fully disclose the conflict of interest to all affected parties and recuse oneself from decision-making roles that create bias
(B) Conceal the conflict
(C) Accept payments from both secretly
(D) Resign from engineering permanently

85. What constitutes 'Falsification' of engineering research data according to international research integrity standards?

(A) Manipulating research materials, equipment, or processes, or changing/omitting data results such that the research is not accurately represented in the research record
(B) Making an honest mathematical error
(C) Refining measurement resolution
(D) Calibrating a sensor

86. In university capstone projects, what does 'Fabrication' of data specifically mean?

(A) Making up data or results and recording or reporting them without performing the physical experiments
(B) Constructing a physical chassis
(C) Soldering a circuit board
(D) Machining a shaft

87. What constitutes 'Plagiarism' under IEEE and university academic honesty policies?

(A) The appropriation of another person's ideas, processes, results, or words without giving appropriate attribution or credit
(B) Collaborating on team projects
(C) Citing referenced textbooks
(D) Reading research papers

88. In safety-critical firmware engineering, why is the use of dynamic memory allocation (`malloc`/`free`) strictly prohibited by MISRA C guidelines?

(A) Dynamic heap allocation introduces non-deterministic execution times, heap fragmentation, and risks catastrophic runtime Out-Of-Memory NULL pointer crashes
(B) Heap allocation uses less RAM
(C) C compilers do not support malloc
(D) Pointers cannot be used in C

89. What is 'Creepage Distance' calculation adjustment in high-altitude substation installations (above 1000 meters)?

(A) Due to lower atmospheric air density and pressure, insulation flashover distances and creepage must be derated by approximately 1.25% per 100 meters above 1000m
(B) Creepage is decreased at high altitude
(C) Altitude has zero effect
(D) Insulators can be shorter

90. In high-voltage SF6 switchgear maintenance, why must SF6 gas recovery and recycling be rigorously enforced?

(A) SF6 is an extremely potent greenhouse gas with a Global Warming Potential (GWP) 23,500 times that of CO2 and an atmospheric lifetime of 3,200 years
(B) SF6 is flammable
(C) SF6 corrodes stainless steel
(D) SF6 is radioactive

91. What safety classification standard (IEC 60529) defines degrees of protection provided by electrical enclosures against dust and water ingress?

(A) IP Code (Ingress Protection, e.g. IP65, IP67)
(B) ISO 9001
(C) IEEE 519
(D) NEMA 1 only

92. What does an enclosure rating of IP67 signify?

(A) Totally dust-tight (6) and protected against the effects of temporary immersion in water up to 1 meter depth for 30 minutes (7)
(B) Explosion proof
(C) Immune to fire
(D) Resistant to 100 kV only

93. What electrical safety standard applies to equipment operated in explosive gas atmospheres (e.g. chemical refineries and mines)?

(A) ATEX / IECEx (Explosion-Proof / Intrinsically Safe Standards, Ex d, Ex i)
(B) UL 94 only
(C) CE mark only
(D) FCC Part 15

94. What is an 'Intrinsically Safe' (Ex i) electrical circuit in hazardous environments?

(A) A circuit designed such that any electrical spark or thermal effect produced under normal or fault conditions is incapable of causing ignition of a specified explosive gas atmosphere
(B) A circuit enclosed in 10-inch steel
(C) A circuit that operates without voltage
(D) A circuit submerged in oil

95. In academic accreditation, what is a 'Course Articulation Matrix'?

(A) A formal mapping matrix demonstrating the correlation levels (Slight, Moderate, Substantial) between Course Outcomes (COs) and Program Outcomes (POs)
(B) A student grade sheet
(C) A faculty timetable
(D) A course fee schedule

96. What is 'Indirect Assessment' in outcome-based education (OBE)?

(A) Assessing student opinions, graduate exit surveys, alumni surveys, and employer satisfaction ratings regarding program outcomes
(B) Final written exams
(C) Laboratory practical exams
(D) Mid-term quizzes

97. What is 'Direct Assessment' in OBE?

(A) Evaluating tangible student coursework performance directly against defined rubric metrics (exam questions, lab records, capstone project rubrics)
(B) Student satisfaction surveys
(C) Alumni feedback
(D) Employer reviews

98. What does the 'Curriculum Advisory Committee' (CAC) or Board of Studies (BOS) in an engineering department do?

(A) Periodically reviews and updates academic curricula with inputs from industry leaders, alumni, and academic experts to align with evolving technologies
(B) Approves student leave
(C) Sets hostel rules
(D) Manages campus catering

99. What is the primary role of a designated Department Safety Officer in electrical engineering facilities?

(A) Conducting safety audits, maintaining PPE inventory, inspecting eye-wash stations, enforcing LOTO protocols, and investigating near-miss incidents
(B) Assigning homework
(C) Booking classrooms
(D) Ordering textbooks

100. What is the ultimate lifelong professional goal instilled by an accredited electrical engineering department?

(A) Empowering graduates to apply rigorous physical sciences and ethical engineering practices to innovate sustainable technologies that benefit humanity
(B) Memorizing equations for exams only
(C) Securing desk jobs without field work
(D) Avoiding complex engineering responsibilities

SECTION B: DESCRIPTIVE & SUBJECTIVE QUESTIONS

Q1. Detail the experimental setup, procedures, and safety precautions of the Electrical Machines Laboratory at LPU.

Q2. Explain the apparatus, testing methodology, and safety interlocks of the High-Voltage and Power Systems Laboratory.

Q3. Detail the equipment, converter topologies, and experimental studies in the Power Electronics & Electric Drives Laboratory.

Q4. Explain the instrumentation, transducers, and feedback control experiments in the Control & Instrumentation Laboratory.

Q5. Analyze the role and activities of professional student societies: IEEE PES, IEEE IAS, and IEEE PELS at university level.

Q6. Detail the Institution of Engineers (India) - IEI: History, student chapters, chartered engineer status, and professional ethics.

Q7. Discuss electrical safety regulations: NFPA 70E, Lockout/Tagout (LOTO), and the Indian Electricity Rules (CEA Regulations 2010).

Q8. Analyze the academic evaluation framework: Continuous Assessment (CA), Course Outcomes (COs), and Program Outcomes (POs).

Q9. Explain the principles and institutional policies regarding Academic Integrity, Anti-Plagiarism, and ethical scientific writing.

Q10. Discuss the design, execution, and evaluation rubrics of undergraduate Capstone Design Projects in Electrical Engineering.

Q11. Explain the test procedure for the measurement of earth resistance using a four-terminal earth ground tester.

Q12. Detail the principles of open-circuit and short-circuit testing on a single-phase transformer to derive its equivalent circuit.

Q13. Describe the operation and calibration of an oil dielectric breakdown tester (BDV test cup) for transformer oil.

Q14. Explain the working and synchronization procedures of an alternator using the Dark Lamp Method.

Q15. Detail the principles and safety rules of operating high-power Variable Frequency Drives in industrial laboratories.

Q16. Explain the construction and working of a synchro error detector pair in position control systems.

Q17. Describe the working principles and calibration of a Pirani Gauge and Penning Gauge for vacuum measurement in switchgear testing.

Q18. Detail the principles of three-phase power measurement using the Two-Wattmeter Method for balanced and unbalanced loads.

Q19. Explain the IEEE Code of Ethics clause regarding health, safety, and welfare of the public with real-world case analysis.

Q20. Detail the principles of continuous quality improvement (CQI) loops and curriculum feedback in engineering education.

Q21. Explain the working and calibration of an optical tachometer for non-contact motor speed measurement.

Q22. Describe the safety requirements for working with high-capacity Lithium-ion battery packs in university research laboratories.

Q23. Explain the structure and functioning of an IEEE Student Branch Executive Committee.

Q24. Detail the design and operation of a Brake Drum Dynamometer for measuring motor mechanical power output.

Q25. Describe the principles of Scott Connection (T-Connection) of transformers for 3-phase to 2-phase conversion.

Q26. Explain the construction and operation of an Earth Leakage Circuit Breaker (ELCB) vs. Residual Current Device (RCD).

Q27. Detail the ethical guidelines for peer review and conflict-of-interest avoidance in scientific engineering publications.

Q28. Explain the function and calibration of a Kelvin Double Bridge for precision low resistance measurement.

Q29. Describe the safety and operational protocols for charging and discharging high-voltage Marx generators.

Q30. Summarize the defining attributes of a modern, ethical, and socially conscious professional Electrical Engineer.

SECTION C: ANSWER KEY - MULTIPLE CHOICE QUESTIONS

Q1:	[A]	**Q21:**	[A]	**Q41:**	[A]	**Q61:**	[A]	**Q81:**	[A]
Q2:	[A]	**Q22:**	[A]	**Q42:**	[A]	**Q62:**	[A]	**Q82:**	[A]
Q3:	[A]	**Q23:**	[A]	**Q43:**	[A]	**Q63:**	[A]	**Q83:**	[A]
Q4:	[A]	**Q24:**	[A]	**Q44:**	[A]	**Q64:**	[A]	**Q84:**	[A]
Q5:	[A]	**Q25:**	[A]	**Q45:**	[A]	**Q65:**	[A]	**Q85:**	[A]
Q6:	[A]	**Q26:**	[A]	**Q46:**	[A]	**Q66:**	[A]	**Q86:**	[A]
Q7:	[A]	**Q27:**	[A]	**Q47:**	[A]	**Q67:**	[A]	**Q87:**	[A]
Q8:	[A]	**Q28:**	[A]	**Q48:**	[A]	**Q68:**	[A]	**Q88:**	[A]
Q9:	[A]	**Q29:**	[A]	**Q49:**	[A]	**Q69:**	[A]	**Q89:**	[A]
Q10:	[A]	**Q30:**	[A]	**Q50:**	[A]	**Q70:**	[A]	**Q90:**	[A]
Q11:	[A]	**Q31:**	[A]	**Q51:**	[A]	**Q71:**	[A]	**Q91:**	[A]
Q12:	[A]	**Q32:**	[A]	**Q52:**	[A]	**Q72:**	[A]	**Q92:**	[A]
Q13:	[A]	**Q33:**	[A]	**Q53:**	[A]	**Q73:**	[A]	**Q93:**	[A]
Q14:	[A]	**Q34:**	[A]	**Q54:**	[A]	**Q74:**	[A]	**Q94:**	[A]
Q15:	[A]	**Q35:**	[A]	**Q55:**	[A]	**Q75:**	[A]	**Q95:**	[A]
Q16:	[A]	**Q36:**	[A]	**Q56:**	[A]	**Q76:**	[A]	**Q96:**	[A]
Q17:	[A]	**Q37:**	[A]	**Q57:**	[A]	**Q77:**	[A]	**Q97:**	[A]
Q18:	[A]	**Q38:**	[A]	**Q58:**	[A]	**Q78:**	[A]	**Q98:**	[A]
Q19:	[A]	**Q39:**	[A]	**Q59:**	[A]	**Q79:**	[A]	**Q99:**	[A]
Q20:	[A]	**Q40:**	[A]	**Q60:**	[A]	**Q80:**	[A]	**Q100:**	[A]

MCQ Scoring Guide & Difficulty Distribution:

Questions 1 to 40: Foundational & Conceptual [Easy] | Questions 41 to 75: Analytical & Applied [Moderate]

Questions 76 to 100: Complex Numerical & Advanced Problem Solving [Difficult]

- Earthing Sticks: Before entering the cage after test completion, technicians must physically touch the high-voltage electrode with an earthing discharge stick to drain residual electrostatic charges.
- Dielectric rubber safety matting (IS 15652) is installed along all operator control consoles.

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1. IEEE Power & Energy Society (IEEE PES) Student Chapter:
Organizes technical seminars on smart grids, grid modernization, and renewable energy. Facilitates participation in the IEEE PES General Meeting and Student Design Competitions, providing access to IEEE Xplore digital library research papers.

2. IEEE Industry Applications Society (IEEE IAS):
Focuses on industrial electrification, motor drives, mining, and factory automation, organizing industrial tours to substations and manufacturing plants.

3. IEEE Power Electronics Society (IEEE PELS):
Conducts hands-on hardware hackathons on EV inverters, wide-bandgap semiconductors, and switch-mode power supplies, cultivating practical hardware prototyping skills.

>>Solution Question Bank: Electrical Safety Engineering (EPAE70) | Lovely Professional University (LPU) | Handwritten Notes, Previous Year Questions (PYQs) & Detailed Solutions 2010).

thresholds ($< 1\,\Omega$), insulation resistance minimums, lightning protection, and mandatory periodic inspection by Electrical Inspectors.

>>Program Outcomes (POs).

smart IoT fault detector, or robotic servo arm).

2. Lifecycle Phases:
- Problem Formulation & Literature Review: Defining measurable specifications, engineering constraints, and environmental/economic viability.
- Hardware/Software Simulation: Prototyping in MATLAB, PSCAD, or Altium.
- Physical Implementation: Fabricating PCBs, assembling switchgear, and coding firmware.
- Validation & Verification: Rigorous laboratory experimental testing against specifications.

3. Evaluation Rubrics:
Evaluated by external and internal panels assessing: Problem Formulation (15%), Methodology & Design (25%), Hardware Fabrication & Testing (30%), Final Report & Documentation (15%), and Oral Defense/Viva (15%).

>>Solution Step 1: Draw the equivalent circuit.

Transformer Oil Dielectric Testing (IEC 60156):

1. Apparatus: Standard test cup (300-500 mL capacity) containing two polished spherical or mushroom-shaped brass electrodes spaced precisely $2.5\text{ mm} \pm 0.05\text{ mm}$ apart.
2. Procedure: The cup is filled with clean oil, allowed to settle for 10 minutes to eliminate air bubbles. Voltage is increased automatically at a uniform rate of 2.0 kV/second until electrical sparkover breakdown occurs. The test is repeated 6 times, and the average breakdown voltage (BDV) is calculated.

>>Solution Store Question Bank and solved papers of several universities Question Papers Solutions of laboratories.

2. Gap Analysis: If student attainment in a specific Program Outcome (e.g., PO5 Modern Tool Usage) falls below target thresholds (e.g. $< 70\%$), the department initiates corrective action.
3. Closing the Loop: The Board of Studies updates syllabi, introduces modern simulation labs (ETAP/PSCAD), or modifies pedagogy, measuring outcomes in the subsequent academic cycle.

>>Solution to Question 24: The establishment of Research Laboratories at Lovely Professional University is driven by the institution's commitment to fostering a culture of innovation, research

Brake Drum Dynamometer:

1. Construction: A cast-iron drum mounted on the motor shaft, enveloped by a friction belt attached to two spring balances (S_1 and S_2). The drum has internal cooling water channels.
2. Mechanical Torque Calculation:
$$T = (S_1 - S_2) \cdot g \cdot R_{eff} \quad [\text{N}\cdot\text{m}]$$
where R_{eff} is effective radius (drum radius + half belt thickness).
3. Shaft Power: $P_{out} = \omega T = \frac{2 \pi N}{60} T \text{ [Watts]}$. Measuring electrical input power yields motor efficiency: $\eta = P_{out} / P_{in}$.

>>Solution Stepwise Question Breakdown: Identify the electrical safety device that trips when it detects a small leakage current to earth, typically 30 mA or less, to protect human life (RCD).